



## Recognising geometric sequences

- The constant multiplier between successive terms in a geometric sequence is called the common \_\_\_\_\_. Fill in the blank
- The common ratio of the geometric sequence 7, 28, 112, 448, 1792, ... is \_\_\_\_\_. Fill in the blank
- 312, 2184, 15 288, 107 016, 749 112, ... is a geometric sequence. Which of these divisions will give you the common ratio? Tick **2** correct answers

☐  $2184 \div 312$

☐  $2184 \div 15288$

☐  $107016 \div 15288$

☐  $107016 \div 749112$

☐  $107016 \div 2184$

- Match each geometric sequence to its common ratio. Write the correct letter in each box

|   |                                 |
|---|---------------------------------|
| a | 16, 80, 400, 2000, ...          |
| b | 4.2, 16.8, 67.2, 268.8, ...     |
| c | 0.07, 0.56, 4.48, 35.84, ...    |
| d | 1602, 4806, 14 418, 43 254, ... |
| e | 20 376, 10 188, 5094, 2547, ... |
| f | 2, 0.4, 0.08, 0.016, ...        |

|  |               |
|--|---------------|
|  | $\frac{1}{2}$ |
|  | 4             |
|  | $\frac{1}{5}$ |
|  | 3             |
|  | 8             |
|  | 5             |



5 The first term of this geometric sequence is \_\_\_\_\_. Fill in the blank

| $n$ | 1 <sup>st</sup>      | 2 <sup>nd</sup>      | 3 <sup>rd</sup> | 4 <sup>th</sup> | 5 <sup>th</sup> |
|-----|----------------------|----------------------|-----------------|-----------------|-----------------|
| $T$ | <input type="text"/> | <input type="text"/> | 2223            | 6669            | 20 007          |

6 What could the third term of this geometric sequence be? Tick 2 correct answers

| $n$ | 1 <sup>st</sup>      | 2 <sup>nd</sup> | 3 <sup>rd</sup>      | 4 <sup>th</sup> | 5 <sup>th</sup>      |
|-----|----------------------|-----------------|----------------------|-----------------|----------------------|
| $T$ | <input type="text"/> | 10              | <input type="text"/> | 40              | <input type="text"/> |

☐ 20

☐ 25

☐ -20

☐ -25

☐ 80

